

HAT-P-54b

R-1828

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_161101 · created 2026-08-03T05:34:22+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457693.85058 +/- 0.00051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2 +/- 0.058
Transit depth (Rp/Rs)^2	3.99 +/- 2.32 [%]
Orbital Inclination (inc)	86.89 +/- 2.33
Airmass coefficient 1 (a1)	1.83 +/- 1.22
Airmass coefficient 2 (a2)	-0.11 +/- 0.52
Scatter in the residuals of the lightcurve fit is	65.26 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	7.0
Transit Duration (day)	0.067 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.2 ± 0.058 vs 0.1572 (deviation 0.74σ)
Duration fit vs published	0.067 d vs 0.075 d (deviation 0.67σ)
Mid-time O–C	-0.7 min
Depth SNR	3.4
Residual scatter	65.26 %

Light curve

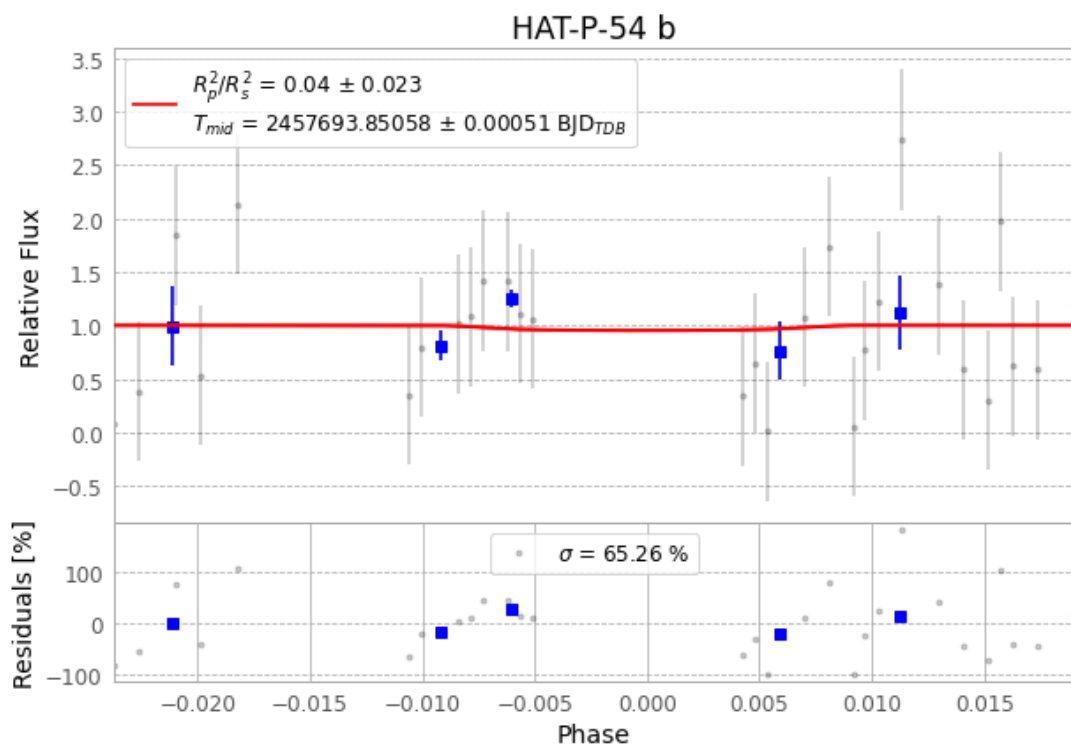


Figure 1 — FinalLightCurve_HAT-P-54 b_2016-10-31.png (SHA-256 2c724251fb78b42e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [535, 463], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 218cd24d7e5f42b6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), HATP-54161101060912.FITS → HATP-54161101111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-161101.log (1924e4243835...), FinalLightCurve_HAT-P-54 b_2016-10-31.png (2c724251fb78...), FinalLightCurve_HAT-P-54 b_2016-10-31.csv (4972d88cd828...)

Compute resources

Wall time 235.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 05:34Z.